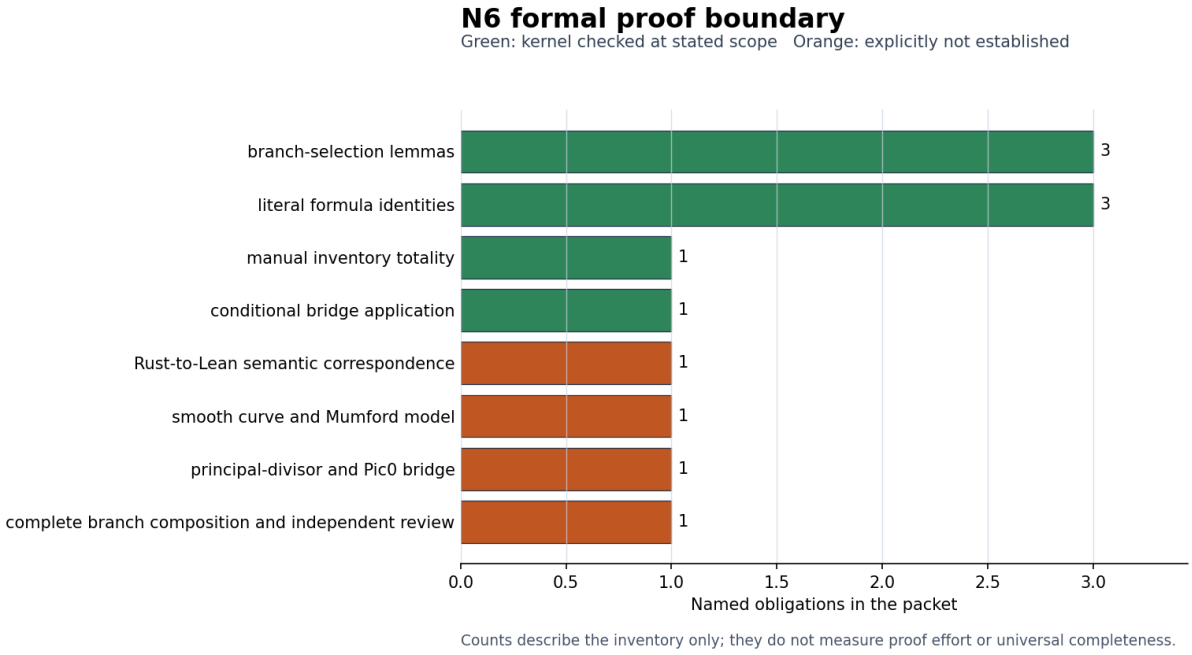
N6 formal proof-boundary report

Partial mechanization • generated 2026-10-25
Schema: `projective-law-formal-report/v1`

Recorded result

Lean 4.34.1 with pinned Mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612` kernel-checks three branch-selection lemmas, three literal formula identities, totality of a manual 17-row reachable-branch mapping, and application of an explicitly caller-supplied divisor-class bridge. The formula identities include exact factorizations for both generic 2+2 interpolation congruences.

The packet does not establish Rust-to-Lean semantic correspondence, a smooth genus-2 curve or Mumford representation in Lean, the principal-divisor and Picard-group bridge, complete exceptional-branch composition, or independent mathematical review. The conditional bridge theorem constructs no bridge and assumes no bridge globally.



Counts are an inventory of named obligations. They do not measure proof effort, importance, or universal coverage.

Standardized source table

Evidence	Status	Scope or limit
Three branch lemmas	Kernel checked	Abstract-field implications with explicit hypotheses.
Generic 1+2 identity	Kernel checked	Literal commutative-ring identity; no divisor argument.
Two generic 2+2 identities	Kernel checked	Exact factorizations by explicit input quadratics.
Runtime mapping	Kernel checked, manual scope	Total over a manually transcribed 17-row inventory; Rust is hash-bound, not parsed by Lean.
Conditional bridge	Kernel checked, conditional	Applies a caller-supplied implication; supplies no geometric bridge.
Source bindings	Integrity checked	Runtime, written audit, and eight native certificate source/result pairs are SHA-256 bound.

Validation and remaining boundary

The cached rebuild completed 740 jobs. The axiom audit found only subsets of `propext`, `Classical.choice`, and `Quot.sound`. The packet verifier and SHA-256 manifest passed.

The next obligation is a faithful formal model joining the hashed Rust case split to reduced Mumford divisor classes on the stated smooth quintic family, including exceptional compositions and degree drops. Until that bridge is supplied and independently reviewed, this packet records algebraic sublemmas and a manual inventory rather than a universal group-law proof.